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DNA Mixtures in Forensic Investigations: The Statistical State of the Art

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Keywords

Bayesian networks, coancestry, DNA databases, deconvolution, DNA mixture, kinship, likelihood ratio, software

Abstract

Forensic science has experienced a period of rapid change because of the tremendous evolution in DNA profiling. Problems of forensic identification from DNA evidence can become extremely challenging, both logically and computationally, in the presence of complicating features, such as in mixed DNA trace evidence. Additional complicating aspects are possible, such as missing data on individuals, heterogeneous populations, and kinship. In such cases, there is considerable uncertainty involved in determining whether or not the DNA of a given individual is actually present in the sample. We begin by giving a brief introduction to the genetic background needed for understanding forensic DNA mixtures, including the artifacts that commonly occur in the DNA amplification process. We then review different methods and software based on qualitative and quantitative information and give details on a quantitative method that uses Bayesian networks as a computational device for efficiently computing likelihoods. This method allows for the possibility of combining evidence from multiple samples to make inference about relationships from DNA mixtures and other more complex scenarios.

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1. INTRODUCTION

Since the pioneering work of Jeffreys et al. (1985), DNA profiling, also known as genetic fingerprinting, has become an indispensable tool for identification of individuals in the investigative and judicial process associated with criminal and civil cases. Civil cases may include paternity and immigration cases, and other contexts (Aitken & Taroni 2004, Balding 2005). Forensic science has experienced a period of rapid change because of the tremendous evolution in DNA profiling. DNA evidence has transformed the proof of identity in litigation, but it has also introduced daunting problems of statistical analysis into the process. Problems of interpretation and identification from DNA evidence can become extremely challenging, both logically and computationally, as the samples are typically of varying quality. These complicating features may include mixed DNA trace evidence, missing data on individuals, heterogeneous populations, kinship, and so on. The features of probabilistic expert systems and Bayesian networks (BNs) have been exploited to handle a wide variety of complex problems of forensic identification, including kinship analysis (Dawid et al. 2002, 2007; Green & Mortera 2009; Corradi & Ricciardi 2013), identification of individuals in DNA mixtures with models based on discrete allelic information (Lauritzen & Mortera 2002, Mortera 2003, Mortera et al. 2003, Biedermann et al. 2012), and continuous peak height information (Cowell et al. 2007a,b, 2011, 2015).

DNA samples found at crime scenes are often complex, either because the amounts of DNA are tiny, consisting of only a few molecules; the samples contain DNA from several individuals; the DNA molecules have degraded; there has been environmental contamination; or a combination of these. In such cases, considerable uncertainty is involved in determining whether or not the DNA of a given individual is actually present in the sample. Such mixed samples contain DNA from multiple individuals. Advanced DNA technology can now extract genetic material from a huge variety of surfaces and objects, like balaclavas and crowbars, which may have been handled by several individuals, only some of whom are related to the specific crime. Therefore, a sample found on the crime scene may contain material from the victim but also from unknown individuals who might be involved in the crime, and determining their DNA becomes an important issue for the criminal investigation.

DNA is now routinely used in criminal investigations and court cases, although DNA samples taken at crime scenes vary in quality and thus present challenging problems for their interpretation. The basic representation of the composition of a DNA sample is the electropherogram (EPG), described further in Section 3.1. The identification of the DNA composition from the EPG information gives rise to a wide range of challenging statistical questions associated with uncertainties and artifacts in the measurement processes, population genetic variations, kinship and coancestry, the number of contributors to the DNA mixture, and estimation of relationships with DNA mixture contributors.

The statistical issues concerning DNA mixtures have been addressed in a number of different ways. Some approaches make use only of the presence or absence of peaks in the EPG or use the peak heights in a semiquantitative fashion; others seek to develop fully quantitative models of observed peak heights. A fully quantitative coherent model that takes all the information in the EPGs into account and gives the uncertainty tied to all the results is the appropriate way to analyze complex DNA mixtures.

The outline of the article is as follows. Section 2 contains a brief introduction to a motivating example, and Section 3 gives an introduction to forensic genetics, such as genetic marker systems, the polymerase chain reaction (PCR) process used for amplifying the DNA, and the artifacts that can occur in this measurement processes. The motivating example is analyzed in Section 4. Section 5 presents an overview of the likelihood or log-likelihood ratio, a method for presenting the weight of the forensic evidence when comparing a prosecution and a defense proposition.

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The loss of evidential efficiency, which indicates how much is lost due to the evidence being based on a mixture rather than a DNA profile from a single contributor, is also described in Section 5. Section 6 presents the various DNA mixture models: those based solely on qualitative information in the DNA mixture, those that use the quantitative peak information without modeling it, and those based on all the qualitative and quantitative information in the DNA mixture. In Section 7, BNs are introduced (brief technical issues on BNs are given in the Appendix). BNs are a computational tool used to solve complex DNA mixtures (in particular, to efficiently represent the genotypes of the unknown individuals) using the model developed by Cowell et al. (2015). This model is presented in Section 8, along with issues concerning computational challenges. Section 9 provides a brief discussion about forensic allele frequency databases and the sensitivity to assumptions about these. The modification of this DNA mixture model for establishing potential relationships, which extends the peak height model to allow for inference about the relationships between contributors to a DNA mixture with unknown genotype and other individuals of known genotype, is introduced in Section 10. In Section 11, we briefly introduce the topic of combining evidence from multiple samples and different marker systems. In Section 12, we illustrate a proposal for computing likelihood ratios for unions of hypotheses. Indications and comparisons between the available open-source software and private commercial software systems used for analyzing DNA mixtures are presented in Section 13. A general discussion and some concluding remarks are given in Section 14.

2. MOTIVATING EXAMPLE: THE POTSDAM MURDER CASE

In 2011, 12-year-old Garrett Phillips was strangled to death in his own home in Potsdam, New York. His murder remains unsolved to this day. The last time Phillips was seen alive was on school surveillance footage, in which he was seen making his way home. Forty minutes later, he was found strangled to death in his mother's bedroom. Oral Nicholas Hillary, the boy's mother's ex-boyfriend, was identified as a suspect based on the analysis of DNA found at the scene.

The New York State Police developed a number of profiles from the fingernail scrapings of the deceased boy. Laboratory analysis showed a DNA mixture with large amounts of DNA (large peak heights) corresponding to the victim, and smaller peaks possibly corresponding to one or more other minor contributors and/or to artifacts in the DNA amplification. Excerpts of the data from the fingernail scraping on two markers (D3S1358 and D19S433) and the victim's genotype are given in Section 4.

The DNA was analyzed independently by two commercial software programs, `TrueAllele`® and `STRmix`™, which gave conflicting results on whether Hillary's DNA was included in the low-level DNA mixture profile (McKinley 2016). `TrueAllele` software gave an exculpatory result, whereas `STRmix` gave an incriminating result for the defendant. `TrueAllele` was used to analyze the fingernail scrapings in 2013 and gave inconclusive results. Nonetheless, based on the incriminating results given by `STRmix`, Hillary was indicted for murder on May 15, 2013. However, in 2016, the judge ruled that the DNA evidence implicating the defendant was inadmissible (*New York v. Oral Nicholas Hillary* 2016) and Hillary was then acquitted.

In Section 4, we show some of the results of the analysis made by the two commercial software systems and by the `DNAmixtures` software (Graversen 2013), highlighting the difficult interpretation of these data.

3. GENETIC BACKGROUND

We now give an overview of the genetic background and specific terminology needed to understand DNA profiling and, in particular, DNA mixtures.

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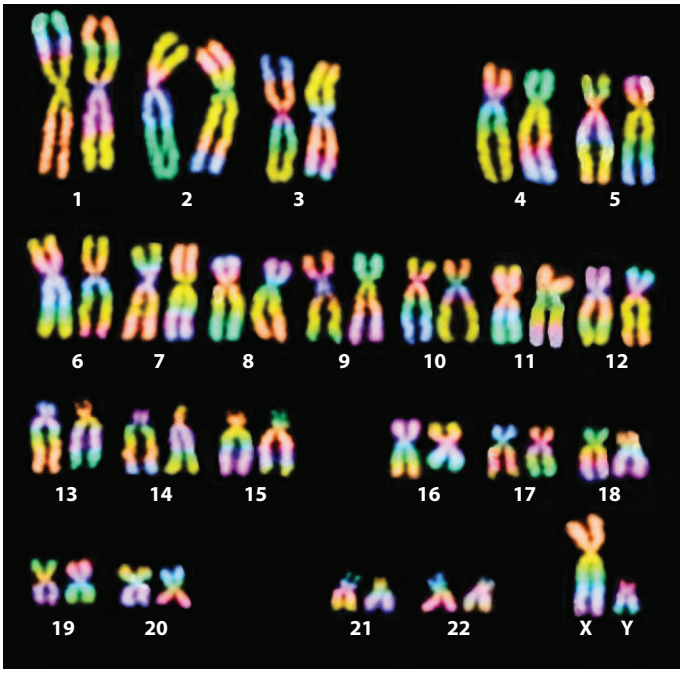


Figure 1

Representation of the 23 pairs of chromosomes in the nucleus of a male human cell.

Forensic scientists encode an individual's genetic profile using the composition of DNA at various positions on the chromosomes. **Figure 1** shows the 23 pairs of chromosomes in the nucleus of a human cell: 22 autosomal homologous chromosome pairs, and a pair that determines sex (male XY, female XX). A chromosome is formed by a double helix composed of the four different nucleotides: cytosine, adenine, guanine, and thymine (C, A, G, and T), binding in pairs C-G and A-T, as shown in **Figure 2**.

A specific position on a chromosome is called a locus. The DNA composition (i.e., the particular sequence of the four nucleotides) on a given locus is an allele. In statistical terms, we can say that a locus corresponds to a random variable and an allele to its realized state. A DNA marker is a known locus where the alleles can be identified in the laboratory.

The information at each locus consists of an unordered pair of alleles that forms the genotype at that locus. It is a pair because autosomal chromosomes come in pairs, one inherited from the father and one from the mother, and unordered because it is not possible to distinguish which allele is inherited maternally and which paternally. Assuming Mendelian segregation, at each marker, a parent passes a copy of just one of his or her two alleles, randomly chosen, to his or her child, independently of the other parent and independently for each child.

A DNA segment is identical by state (IBS) in two or more individuals if they have identical nucleotide sequences in this segment. An IBS segment is identical by descent (IBD) in two or more individuals if they have inherited it from a common ancestor without recombination, i.e., the segment has the same ancestral origin in these individuals.

When the population is in Hardy–Weinberg equilibrium, it implies that the two alleles of an individual can be assumed to be sampled independently from the same population. For forensic markers, we can assume independence of alleles within and across markers, i.e., Hardy–Weinberg

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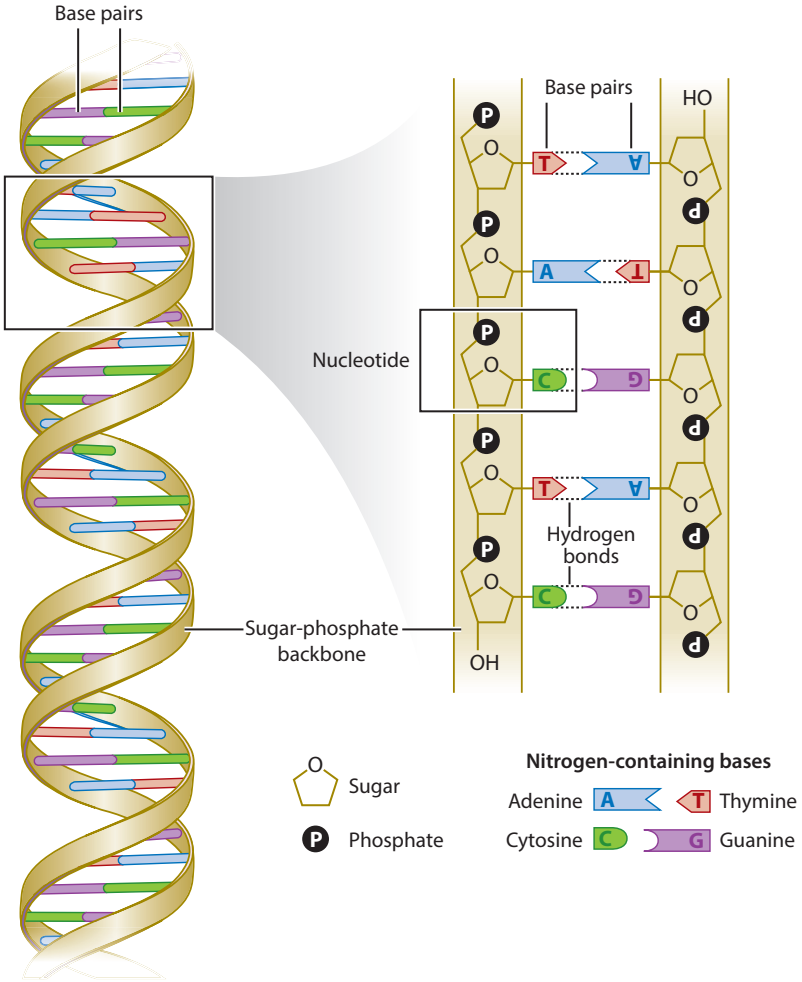


Figure 2

The DNA double helix is composed of 4 different nucleotides: C, A, G, and T, binding in pairs C-G and A-T.

and linkage equilibrium, as they are located on different chromosomes (or far apart if on the same one).

The loci used for forensic identification have been chosen for various reasons. First, the loci chosen are highly polymorphic, so that there is great variability among individuals in the alleles that may be observed. This variability can therefore be exploited to differentiate among individuals. Second, the alleles at each locus can be considered independent. Furthermore, these loci are noncoding, so they are not subject to evolutionary pressures, sexual selection, and related factors that could interfere with the assumed independencies. This simplifies statistical analysis.

Short tandem repeat (STR) markers, also known as microsatellites, are the alleles most frequently used in forensic genetics. Single-nucleotide polymorphisms (SNPs), defined as DNA sequence variations that occur when a single nucleotide (A, T, C, or G) in the genome sequence is altered, are now used sometimes. An STR allele is typically labeled by its repeat number, usually

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...AATGAATGAATGAATGAATGAATGAATGAATG...

Figure 3

An allele with 8 repeats of the 4 nucleotides AATG, designated by $[AATG]_8$.

an integer. For example, consider marker TH01 with allele having repeat number 8. This allele includes the sequence of four nucleotides (a tetranucleotide) AATG repeated consecutively eight times. It is designated by the formula $[AATG]_8$ and can be represented as in **Figure 3**.

Repeat numbers are not always integers. For example, allele 8.3 of TH01 corresponds to $[AATG]_5ATG[AATG]_3$, in which 8 refers to the eight complete four-word bases $[AATG]$ and .3 refers to the three base-long word sequence ATG in the middle. Repeat numbers with decimal .1 and .2 endings are also possible, indicating the presence of a word of one or two bases. Some markers, e.g., D22S1045, are based on sequences of three nucleotide (trinucleotide) motifs rather than tetranucleotides as above. The base-letter sequences for many alleles may be found in Butler (2005).

When two alleles of an individual at a locus are the same, then the individual's genotype for that locus is said to be homozygous; when the repeat numbers differ, the genotype for that locus is said to be heterozygous. An actor's DNA profile is typically a list of genotypes at 16–22 markers. It is customary to assume that all actors are drawn from a homogeneous population in Hardy–Weinberg equilibrium, with known allele frequencies in some reference database. Green & Mortera (2009) show how these assumptions can be relaxed. Results showing the sensitivity to these assumptions are given in Section 9.

A haplotype is a group of alleles in an organism that are inherited together from a single parent, as occurs for the male Y-chromosome. A haplotype can also refer to specific DNA sequences in a cluster of tightly linked genes on a chromosome that are likely to be inherited together, i.e., they are likely to be conserved as a sequence for many generations.

3.1. Genetic Markers Used for DNA Profiles

The main detection technology for genetic markers used for forensic purposes is based on STR markers. Other genetic marker systems, including SNPs and insertions and deletions of autosomes and Y-chromosomes, are currently being developed. One problem is that criminal databases used for forensic investigations are based on STR markers, so the use of SNPs (Voskoboinik et al. 2015) and recent technologies is still premature, as they would require establishment of SNP databases to combine with existing STR information.

A special type of haplotype is the Y-chromosome STR, which can be used to detect male contributors to a mixture. More general use of haplotypes in forensic investigation of mixtures however, is limited by the lack of statistical methods able to address the many challenges that arise with haplotype DNA mixtures (Andersen et al. 2015). For familial identification and inference about maternal ancestry, mitochondrial DNA (mtDNA) is used. The rapidly developing massive parallel sequencing (MPS) technology may be extremely useful for analyzing DNA mixtures in the future (Hwa et al. 2018). Other emerging markers, including DIP-STRs (a pairing of deletion-insertion polymorphisms, with STR markers) have the potential to become more widely used.

3.2. The Polymerase Chain Reaction Process

The DNA collected from a crime scene for forensic analysis consists of a number of cells from one or more individuals. Note that most human cells contain two alleles (diploid cells) for each

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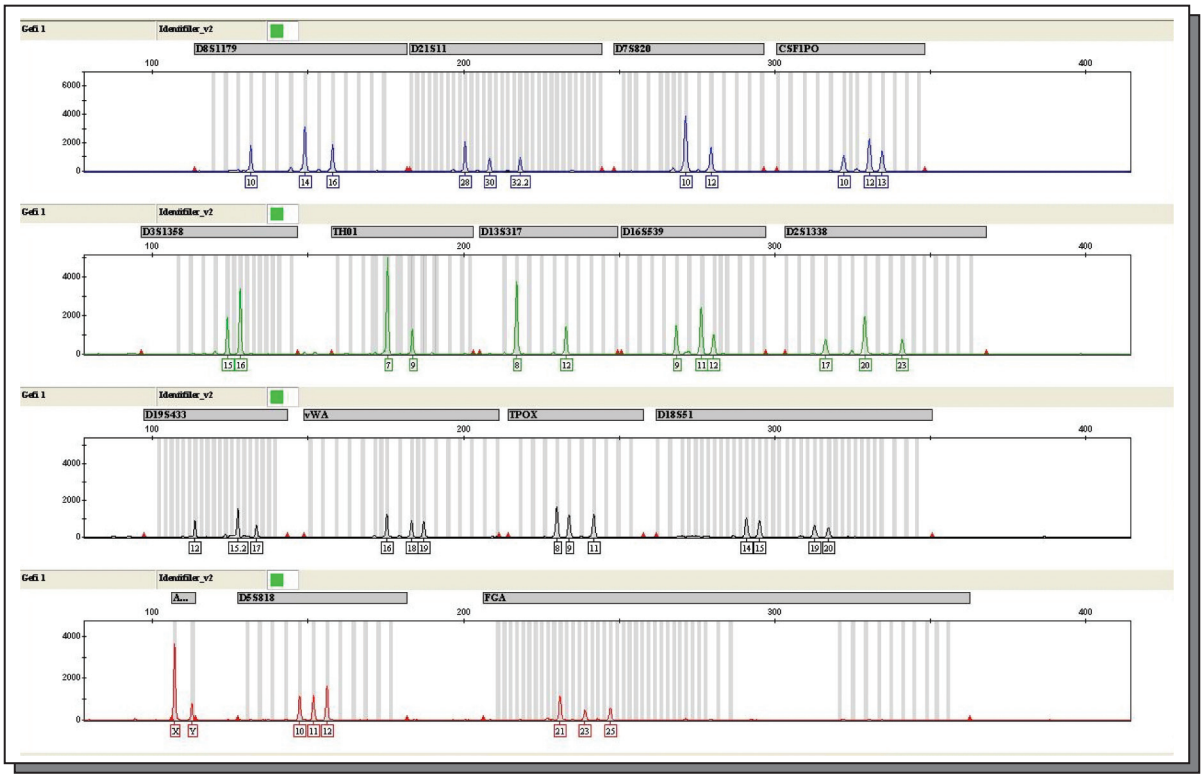


Figure 4

An EPG using the Identifiler™ STR kit. Each of the four panels represents a dye. Marker names are in gray boxes above panels and repeat numbers of detected alleles are given below the corresponding peaks. The vertical gray bars are used to identify alleles by their length in base pairs. On the left-hand side of each panel, the scale in RFU is indicated. Abbreviations: EPG, electropherogram; RFU, relative fluorescence units; STR, short tandem repeat.

autosomal marker, except gamete cells, which have only one allele (haploid cells). This means that in a mixture, a particular individual will contribute the same number of alleles for each marker.

The DNA in the extract is then amplified using the PCR process. This involves adding primers and other biochemicals to the extract and then subjecting it to a number of rapid heating and cooling cycles. Mathematically, the amplification process may be modeled as a branching process (Sun 1995). The amplification process is not 100% efficient—not every allele gets copied in each cycle—so if two distinct alleles in a marker are present in the extract in the same amount prior to amplification, they will occur in different amounts at the end of the PCR process. When there is sufficient DNA in the sample to carry out PCR amplifications with several extracts, one obtains replicates.

The data may be presented to a forensic scientist as an EPG, as shown in **Figure 4**, with each panel in the EPG corresponding to a different DNA-binding dye. The horizontal axis indicates the base pair length, and the vertical axis shows the intensity. In the absence of artifacts, a peak in the EPG indicates the presence of an allele in the sample before amplification. The peak height is a measure of the amount of the allele in the amplified sample expressed in relative fluorescence units (RFU). Both peak height and peak area are measured by software in the detecting apparatus.

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We refer to the alleles and peak size information extracted from the EPG as the profile of the DNA sample, or more briefly, the DNA profile. Commonly, a DNA profile is a list of genotypes at 16-22 STR markers.

3.3. Artifacts

We now give a brief description of the main artifacts that commonly occur in the DNA amplification process.

3.3.1. Missing peaks. When peaks are not amplified or are too small to appear above a detection threshold, a dropout occurs. In this case, there is too little DNA to detect a peak raise above a minimum threshold.

In measuring the peak heights, low-level noise gives rise to small spurious peaks. A peak amplitude threshold C may be set by the forensic analyst, below which peaks are ignored. Thus, an allele present in the DNA sample will not be recorded as observed if the peak it generates is below the threshold; when this happens, a dropout of the allele is said to have occurred. A dropout can also occur simply by sporadic failure of the apparatus. Dropout is an artifact that can make the analysis of both mixed and single-source DNA samples difficult.

3.3.2. Stutter. Another common artifact is stutter, in which an allele that is present in the sample is miscopied at some stage in the PCR amplification process, and a four-base-pair word segment is omitted. This damaged copy itself takes part in the amplification process, and so yields a peak four base pairs below the allele from which it arose. The most common stutter peak occurs when the PCR process gives copies one repeat too short, forming a back stutter. **Figure 5** shows the loss of one repeat AGAT in the PCR amplification, which in turn will be amplified so some alleles 7 will appears as 6 on the EPG. More rarely, other stutter patterns occur, like a back stutter two repeats too short, or a forward stutter one repeat longer.

3.3.3. Dropin. Another artifact is known as dropin, referring to the occurrence of small unexpected peaks in the EPG caused by contamination by small amounts of DNA irrelevant to the case. This can be due, for example, to sporadic contamination of a sample either at the source or in the forensic laboratory.

Current technology allows for the amplification of very little DNA material, even as little as is contained within one cell, sometimes coming from touch DNA. Amplification of small amounts of DNA is known as low template DNA (LTD). LTD is particularly prone to artifacts such as dropin and dropout.

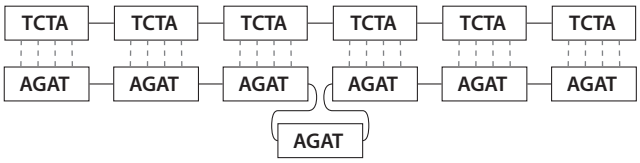


Figure 5

A pictorial representation of a back stutter artifact, in which an allele present in a DNA sample is miscopied at some stage in the PCR amplification process. The loss of one repeat AGAT in the PCR amplification shows how an allele with repeat 7 will appear as allele 6 on the EPG. Abbreviations: EPG, electropherogram; PCR, polymerase chain reaction.

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3.3.4. Silent alleles. A mutation in the flanking region can result in the allele not being picked up at all by the PCR process, in which case we say that the allele is silent. An allele can also be undetectable, and thus de facto silent, because its length is off-scale and the peak therefore does not appear in the EPG.

3.3.5. Pull-up alleles. Pull-up is an artifact whereby the signal (a high peak) in one dye lane might bleed over (i.e., pull-up) into another dye lane in the same position. For example, a signal from a locus in the blue dye might mistakenly be interpreted as a yellow or green signal, thereby creating false peaks at the yellow or green loci.

3.3.6. Warning against purely subjective evaluation of electropherograms. Some methods, like random man not excluded (RMNE) or combined probability of inclusion (CPI) (discussed further in Section 6.1), rely also on a visual examination of the EPG. Simple visual examination of EPGs can lead to misleading analyses. Take, as an extreme case, the example of the DNA profile on one marker shown in **Figure 6**.

A visual examination of **Figure 6** might adopt one of the following explanations for the peaks:

- A single source with genotype (18, 18) and stutter at 17
- A single source with allele 18, stutter at 17, and a dropped-out allele
- A single source with genotype (17, 18) and a large amount of heterozygous imbalance
- A two-person mixed trace with genotypes (17, 17) and (18, 18) in proportions (0.25, 0.75)
- A two-person mixed trace with genotypes (17, 18) and (18, 18) in proportions (0.50, 0.5)

We see that visual examination gives a mixed message that could lead to many possible explanations. The statistical analysis might also give a mixed message, but the weights on the different interpretations are objectively defined. Thus, statistical models are an essential tool to analyze DNA mixtures and give a measure of the uncertainty in the results. Some years ago, Evett et al. (1998) suggested the need for probabilistic genotyping models, stating that expert systems technology might be the most profitable path forward.

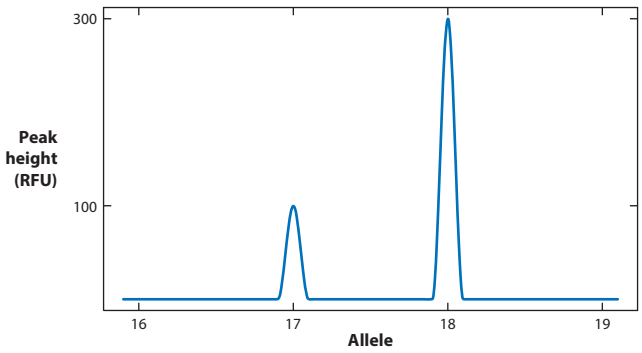


Figure 6

Example of a DNA profile on one marker. How was this profile formed? There are multiple potential explanations based on visual examination, but statistical analysis provides more informative results. Abbreviation: RFU, relative fluorescence units.

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Table 1 Alleles and peak heights in the DNA mixture and the victim's genotype for an excerpt of the markers in the Potsdam murder case

Marker	Allele	Peak height (RFU)	Victim
D3S1358	13	241	
	14	*4,149*	14
	15	233	
	16	*3,770*	16
	17	84	
	18	31	
D19S433	11	80	
	12	225	
	13	*3,431*	13
	13.2	49	
	14	253	
	15	*3,375*	15
	18.2	55	

Peak heights (RFU) in correspondence with the victim's genotype are shown in italic. Abbreviation: RFU, relative fluorescence units.

4. ANALYSIS OF POTSDAM MURDER CASE

In Section 2, we introduced the controversial murder case involving a DNA mixture, commonly referred to as the Potsdam murder case.

The data from the fingernail scrapings for two markers[1] are shown in **Table 1**. It gives the alleles present in the mixture, the corresponding peak heights, and the genotype of the victim. One can immediately see that the peaks in correspondence with the victim's genotype (italicized entries) are extremely high compared with the other peak heights.

On the basis of these data, several different interpretations can be made.

- The number of contributors to the mixture could be two, three, or more, some perhaps due to contamination.
- If there were only two contributors, then on marker D3S1358, the peaks with smaller heights at positions 13 and 15, associated with major peaks at 14 and 16, need to be interpreted as stutter peaks.
- If there were three or more contributors, alleles 13 and 15 could represent a true allele from a minor contributor, or they could be a stutter peak from a neighboring allele or the cumulative peak from both stutter and a minor contributor.
- If a threshold of C = 50 RFU is used, then the peak at allele 18 on D3S1358 with height of 31 RFU, and the peak at allele 13.2 for D19S433 with peak height of 49 RFU, would be excluded from the analysis.

In the analysis made by `STRmix`, a threshold of 50 RFU was used, thus excluding the possibility that allele 18 at D3S1358 and at 13.2 for D19S433 are true alleles. Furthermore, the two peaks in stutter position were probably defined as stutter peaks and not considered in the analysis. In a commentary on the case, Perlin (2016) states that "the fingernail data contains potentially exculpatory DNA peaks between 30 and 50 rfu that do not come from the defendant. According to `STRmix`, the mixture ratio is around 250:1." A mixture ratio of 250:1 implies that the proportion contributed to the mixture by the victim is about 99.6%, and the remaining 0.4% is contributed by

[1]We were not able to obtain the full data from this case, only that which appears in **https://johnbuckleton.files.wordpress.com/2017/12/people-v-hillary-ii.pdf**.

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Table 2 Results for an excerpt of the markers in the Potsdam murder case

		DNAmixtures		TrueAllele		STRmix	
Marker	Rank	Genotype	Prob.	Genotype	Prob.	Genotype	Prob.
D3S1358	1	16,17	0.190	16,17	0.35	14,17	0.34
	2	15,17	0.171	14,16	0.24	16,17	0.28
	3	14,17	0.146	14,17	0.20	17,17	0.16
D19S433	1	12,14	0.214	12,14	0.85	11,11	0.49
	2	12,13	0.160	14,14	0.08	11,13	0.21
	3	13,14	0.118	13,14	0.05	11,15	0.19

The results we obtained when using DNAmixtures are based solely on the two markers in **Table 1**, whereas those for TrueAllele and STRmix are based on data on all the markers and possibly on different runs. Abbreviation: Prob., probability.

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an unknown individual. Numerous artifacts, such as dropout alleles, are customary when a DNA mixture has such a large peak imbalance between the contributors. If not analyzed appropriately and extremely carefully, a complex DNA mixture with a large imbalance, as in this case, can lead to the incorrect identification of the donor.

One of the developers of STRmix posted a statement online about the case, including that the most plausible explanation for the peak at 18.2 on D19S433 is that it is an artifact due to pull-up, as it is in the same position as an allele having a very large peak height of 3281 RFU in another dye lane (**https://johnbuckleton.files.wordpress.com/2017/12/people-v-hillary-ii.pdf**).

Table 2 shows the results on the two markers in **Table 1** when predicting the 3 most probable genotypes of the unknown contributor. In the third and fourth columns, the genotypes and corresponding predictive probability were estimated using DNAmixtures, but based solely on the data for the two markers (the full data were not made available to the author); the fifth and sixth columns show the results for TrueAllele (a rough estimate of the probabilities is given, as these were read from a graphical representation); and the last two columns give the results with STRmix. The latter two are based on the complete data on many markers and possibly many runs. Note that even with the limited amount of data available, the results using DNAmixtures are quite similar to those using TrueAllele, which lead to an exculpatory result for the defendant. The results we obtained using DNAmixtures, based on the limited data available, give higher uncertainty on the predicted genotypes, as expected.

There is a remarkable difference between the results from TrueAllele and DNAmixtures compared with those given by STRmix in the top ranking predicted genotype of the minor donor for marker D19S433. The three top ranking genotypes predicted by STRmix for this marker are completely different from those predicted with the other two models.

Recall that the results given by the analysis made with STRmix led to Hillary being indicted for murder. Ultimately, the judge did not admit the results based on the probabilistic DNA evidence (*New York v. Oral Nicholas Hillary* 2016).

5. THE WEIGHT OF EVIDENCE

In this section, we discuss approaches to present the results of a forensic analysis. The weight of the evidence when comparing a prosecution and a defense proposition we believe should be defined as a likelihood ratio (LR) (see Section 5.1), it being the responsibility of the trier of fact to combine the LR appropriately with his or her prior assessment, in order to compute the posterior probability of the prosecution and defense hypotheses. In the absence of a suspect, investigation

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may begin by identifying the most plausible genotypes of the unknown perpetrator(s). This can be achieved by deconvolving the DNA mixture into a list of the most likely unknown contributor's genotypes and calculating the profile probability of each. The results shown in **Table 2** were produced by deconvolving the DNA mixture.

5.1. Evidential Calculations

The available evidence E consists of the peak heights as observed in the EPG as well as the genotypes of any known individuals. Let E denote one or more items of evidence (perhaps its totality). We need to consider how this evidence affects the comparison of the prosecution hypotheses $\mathcal{H}_p$ and the defense hypothesis $\mathcal{H}_d$, offered by the two sides.

For example, $\mathcal{H}_p$ could be that the suspect s, the victim v, and an unknown U contributed to a mixture profile. Under $\mathcal{H}_p$, s and v are known contributors. Alternatively, $\mathcal{H}_p$ could involve s and v as known contributors, in addition to one or more unknown individuals who we call $U_1, U_2, U_3, \ldots$ and refer to as unknown contributors. The alleles of the unknown contributors are usually assumed to be chosen randomly and independently from a reference population with known allele frequencies. The prosecution hypothesis is then compared with the defense hypothesis $\mathcal{H}_d$, which typically would replace the suspect s in $\mathcal{H}_p$ with an unknown contributor, claiming that the apparent similarity between the trace and the DNA profile of s is due to chance. (The defense hypothesis is not actually advocated by the defense, but is formulated strictly for comparative purposes.)

The strength of the evidence (Good 1950, Lindley 1977) is normally represented by the likelihood ratio:

$$LR = \frac{L(\mathcal{H}_p)}{L(\mathcal{H}_d)} = \frac{\Pr(E \mid \mathcal{H}_p)}{\Pr(E \mid \mathcal{H}_d)}. \quad 1.$$

The weight of evidence $WoE = \log_{10} LR$, in units of ban was introduced by Alan Turing (Good 1979) so that one ban represents a factor 10 on the LR. For a high-quality, single-source DNA profile (i.e., from a single individual), values for WoE vary around 18 bans, which in most cases would be considered sufficient for establishing the identity of the individual leaving the stain. We emphasize that this unit of WoE should always be interpreted relative to specific prosecution and defense hypotheses.

The calculation of the WoE can be important both when presenting a case to the court and in the investigative phase of a trial when deciding whether it is worthwhile to search for additional independent evidence.

Given the evidence E based on the alleles and peak heights in the DNA mixture profile, the numerator and denominator in the LR are calculated based on models that generally have the form

$$\Pr(E \mid \mathcal{H}) = \sum_{\mathbf{g}} \Pr(E \mid \mathbf{g}) \Pr(\mathbf{g} \mid \mathcal{H}), \quad 2.$$

where the model for the conditional distribution $\Pr(E \mid \mathbf{g})$ of the evidence given the genotypes $\mathbf{g}$ of all contributors is the same for both hypotheses, whereas the distribution $\Pr(\mathbf{g} \mid \mathcal{H})$ of genotypes of the contributors differs according to the hypothesis $\mathcal{H}$.

5.2. Deconvolution of DNA Mixtures

The deconvolution of a DNA mixture attempts to identify the combined genotypes across all markers of each unknown contributor to the mixture and give a list of potential genotypes of a

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perpetrator to use for investigative purposes. For example, we could wish to calculate

$$\Pr\{U_1, U_2 \mid E, \mathcal{H}^*\} \text{ or } \Pr\{U_1 \mid E, \mathcal{H}^*\},$$

where U_1, U_2 represent genotypes of unknown contributors to the mixture under an investigative hypothesis $\mathcal{H}^*$, such as, the trace contains DNA from three persons, one of whom is the victim v, or, for example, two of whom are v and s.

5.3. Evidential Efficiency

For a single-source, high-quality trace matching the suspect s, the WoE is simply $-\log_{10}\pi_s$, where $\pi_s = \mathrm{P}(U = s)$ is the match probability, i.e., the probability that a random member of the population has the specific DNA profile K_s of s. We point out that the WoE against a suspect based on a mixture model can never be stronger than the WoE obtained by a single-source DNA profile. Let $\mathcal{H}_p$ be the prosecution hypothesis involving s as a contributor and $\mathcal{H}_d$ the defense hypothesis, replacing the genotype s by that of a random individual U. One can show that

$$LR = \frac{\Pr(E \mid \mathcal{H}_p)}{\Pr(E \mid \mathcal{H}_d)} \leq \frac{1}{\pi_s}. \quad 3.$$

Thus, we have $WoE = \log_{10} LR \leq -\log_{10}\pi_s = WoE(K_s)_{\max}$, implying that a mixed trace can never give stronger evidence than a high-quality trace from a single source.

The loss of evidential efficiency $\mathrm{WL}(E \mid K_s)$ against the suspect having genotype K_s for evidence E can be defined as

$$\mathrm{WL}(E \mid K_s) = WoE(K_s)_{\max} - WoE(K_s) = -\log_{10}\pi_s - \log_{10}\frac{\Pr(E \mid \mathcal{H}_p)}{\Pr(E \mid \mathcal{H}_d)}.$$

This quantity indicates how many bans of WoE are lost due to the evidence being based on a mixture rather than a single-source trace.

In the case of deconvolution without a specified suspect, we could consider the evidence against the most likely unknown person K^* under the defense hypothesis. If $\mathcal{H}_p^*$ denotes a prosecution hypothesis obtained by replacing U with K^* and $\pi^* = \mathrm{P}(U = K^*) = \mathrm{P}(U = K^* \mid \mathcal{H}_d)$ denotes the prior probability of a random individual having genotype K^*, based on the database allele frequencies, one can show that

$$\mathrm{WL}(E \mid K^*) = -\log_{10}\Pr(U = K^* \mid \mathcal{H}_d, E). \quad 4.$$

Thus, the generic loss of evidential efficiency is equal to $-\log_{10}$ of the maximum posterior probability obtained in the deconvolution. For a single-source trace that uniquely identifies the contributor, $\mathrm{WL}(E \mid K^*) = 0$.

The loss of evidential efficiency is particularly relevant for assessing the value of DNA evidence in the investigative phase or prior to a courtroom presentation.

6. DNA MIXTURE MODELS

Methods for the interpretation of DNA profiles arising from mixtures can be classified into:

1. those based on qualitative information, i.e., information about allele presence or absence in the DNA mixture, termed binary methods;

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2. those based on qualitative information that use but do not model the peak amplitude information, termed semicontinuous models; and
3. those that model the quantitative information by taking into account both allele presence and peak intensities, termed continuous or probabilistic genotyping methods.

We strongly believe that method 3 is the best and most coherent way to model DNA mixtures as it takes all the evidence available in one or more DNA mixture samples into account.

We first discuss some methods that are based prevalently on allele presence, and then we describe those that also exploit the information in the peak intensities. A list of publications relating to DNA mixtures is maintained at the website of the National Institute of Standards and Technology (NIST) (**http://www.cstl.nist.gov/strbase/mixture.htm**).

6.1. Methods Based on Qualitative Information

We first describe the most common methods that are based solely on allele presence, and then we illustrate methods that also include some information on peak heights, without directly modeling these.

6.1.1. Combined probability of inclusion or random man not excluded. The CPI or RMNE method estimates the proportion of unrelated individuals randomly picked out of a general population who would not be excluded as possible contributors to a DNA profile. A probability of inclusion (PI) is calculated at each locus by adding the population frequencies of the alleles greater than a threshold and squaring their sum.

Multiplying the PI values of included loci produces the standard CPI match statistic for the mixture. The CPI or RMNE approach omits any knowledge on the number of contributors and is regarded as less powerful and informative than the LR approach. The inclusion probability for DNA mixtures is a subjective method unrelated to identification information. A further danger in using the CPI method is that increasing the number of STR loci generally yields a higher CPI value regardless of their informativeness.

BNs can be used to structure and solve complex cases of forensic identification involving DNA traces, based solely on allele presence, that might be mixtures of several DNA profiles. This approach was used to handle cases where the number of contributors to the mixture could not be regarded as known in advance or where the mixed DNA trace was compounded by such features as missing individuals or the possibility of unobserved alleles in the trace (Mortera et al. 2003).

6.1.2. Semicontinuous models. These methods are based on the analysis of allelic presence but also include some information on peak heights. They consider a single DNA marker and, at that marker, allow for artifacts such as dropin and dropout, including the probability of their occurrence in the LR computations. Under the independence assumptions of this model, the authors state that the model readily extends to multiple loci and DNA mixtures (Curran et al. 2005, Balding & Buckleton 2009, Gill et al. 2012).

Curran et al. (2005) give a set-based method for LR calculations, which includes multiple contributors and the analysis of replicate runs. This was an improvement on previous guidelines for the analysis of LTD profiles where alleles were not reported unless they were duplicated in replicate runs (Gill et al. 2000). They consider the probability of contamination and dropout, but not of stutter. The model is implemented in Gill et al. (2007). For the interpretation of LTD, Gill et al. (2008) include the probabilities of dropout, contamination, and stutter; peaks in stutter positions are considered ambiguous alleles and included in the calculation.

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Balding & Buckleton (2009) consider a discrete model for interpreting LTD mixtures where peaks are classified as present, absent, or masking. The set of masking alleles is defined as the collection of peaks that exceed a designated threshold and that either correspond to an allele of a known contributor or are in a stutter position to that allele or another peak of sufficient height. Based on an extension of that paper, Balding (2013) wrote a suite of R functions `likeLTD` built for a range of crime scene DNA profiles involving complex mixtures, uncertain allele designations, dropin and dropout, degradation, stutter, relatedness of alternative possible contributors, and replicated runs. Alleles are defined as present, absent, or uncertain; dropout is modeled using the method in Tvedebrink et al. (2009, 2012). Haned & Gill (2011) estimate dropout probabilities using different logistic models for haploid and diploid cells in combination with a simulation model of the PCR process (Gill et al. 2005). Similarly, Haned et al. (2012) and Haned & Gill (2011) adopt a model based on allele presence that analyzes the sensitivity to dropin and dropout parameters. The papers mentioned in this paragraph do not make direct use of peak heights, but peak heights enter implicitly in the process of allele classification.

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6.2. Methods Based on Quantitative Information

Early attempts to exploit peak intensity information include those of Evett et al. (1998, Perlin & Szabady (2001) and Bill et al. (2005). Cowell et al. (2007a) introduced the use of gamma distributions to model peak area variability observed in the PCR amplification of DNA in mixtures and presented a normal-approximation version of the model in Cowell et al. (2007b). These papers used exact probability propagation algorithms for BNs but required discretization of the parameter representing the relative amounts of DNA from each contributor to the mixture. Both evidential and deconvolution analyses were presented. The suitability of the gamma distribution was investigated by Cowell (2009) using the simulation model of Gill et al. (2005).

The gamma model of Cowell et al. (2007a) was extended by Cowell et al. (2011) to handle artifacts such as stutter, dropout, and silent alleles. The authors showed how the BN representation readily enabled the simultaneous analysis of two DNA mixtures that are thought to have contributors in common. The authors noted that the computational complexity of the model increased significantly with the total number of contributors, limiting the practical use of their methods to, at most, four contributors.

Perlin (2011) presented a model for DNA mixture interpretation based on a normal model in a direction similar to Cowell et al. (2007b). They used a fully Bayesian approach where all parameters are given a prior distribution. The model is used for both identification of contributors to a DNA mixture and deconvolution. The model does not incorporate stutter, and dropout is accounted for by a background variance parameter. The performance of the model is reported for 16 two-person mixture samples; however, the performance for mixtures of DNA from more than two people is not reported.

The work of Puch-Solis et al. (2013) is based on using the gamma distribution for peak heights, as by Cowell et al. (2007a, 2011). Since DNA degrades more with increasing DNA fragment length (Tvedebrink et al. 2012), they use a correction for this phenomenon. Their methodology is different from that of Cowell et al. (2015) in that (*a*) they exploit the common scale parameter for peak heights to derive Dirichlet distributions for relative peak heights, (*b*) they use a preprocessing step for classifying peaks as possible stutter peaks, (*c*) they use a more complex dropout model, (*d*) they discretize the parameter representing the proportion of DNA from each contributor and use a finer grid on extreme intervals to capture better unbalanced mixtures, and (*e*) all alleles not corresponding to peaks above threshold are lumped together into a single compound allele. They discuss identification of contributors to the mixture and not deconvolution. The

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paper presents examples of two-person mixtures and does not analyze replicate runs or multiple traces.

Tvedebrink et al. (2010) evaluate the weight of evidence for two-person mixtures using a multivariate normal distribution of peak heights. Based on DNA mixtures from controlled experiments, they find a linear relationship between peak height and area and between means and variances of peak height measurements. Controlled experiments are also used for estimating the probability of allelic dropout per locus using logistic regression (Tvedebrink et al. 2009, 2012). Bright et al. (2013) use a log-normal model for the ratio between observed and expected peak heights.

7. BAYESIAN NETWORKS AND CONDITIONAL PROBABILITY TABLES

The main computational challenge in DNA mixtures is associated with the marginalization over all unknown DNA profiles, as in Equation 2, needed for evaluating the likelihood function. To enable efficient computation, an appropriate BN is used to represent the genotypes of the unknown contributors and the observations. BNs, with inferences computed by probability propagation methods (junction tree algorithms), offer an appealing practical modeling framework for structured systems involving discrete variables in numerous domains, including forensic genetics, and in the following, we make extensive use of such networks for model specification and computation of inferences. Here, we present the details that are needed to comprehend the use of BNs in DNA mixture models. Some further features of BNs are summarized in the Appendix. For a detailed account of the theory of BNs, the reader is referred to Cowell et al. (1999).

A BN is a form of directed acyclic graph (DAG) comprising a finite set V of nodes, with arrows between some of the nodes in such a way that it is not possible, by following the arrows, to return to one's starting point. An example is shown in **Figure 7**. Nodes with arrows pointing out from them and into a node $i \in V$ are called its parents; those with arrows pointing into them out from i are its children. In a BN model, the joint distribution of all variables factorizes as a product of parent-to-child conditional distributions:

$$p(x) = \prod_{i \in V} p(x_i | x_{\mathrm{pa}(i)}),$$

where $x_{\mathrm{pa}(i)}$ denotes the parents of i.

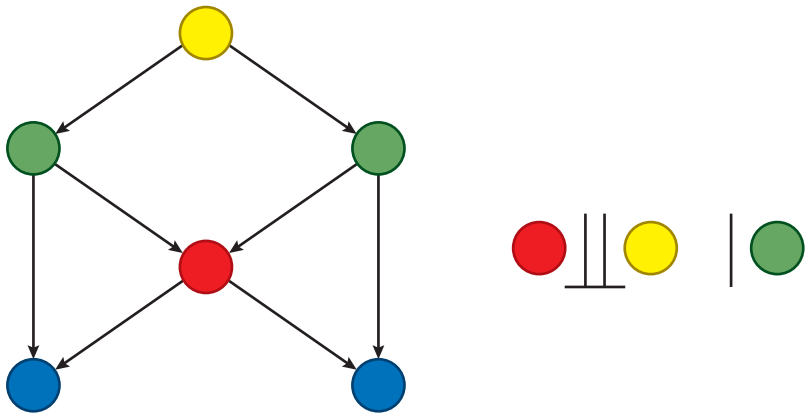


Figure 7

A Bayesian network (BN). The green nodes have the red node as their child, and it, in turn, is a parent to the blue nodes. The notation on the right states that the red node is independent of the yellow node, conditional on the green nodes.

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Figure 7 shows a BN where, the green nodes have the red node as their child, and it is in turn a parent node to the blue nodes. Conditional independences among the nodes can be read from a BN. For example, the bottom right notation in the figure (Dawid 1979) states that the red node is independent of the yellow node, conditional on the green nodes (parents of the red node).

When variables are discrete, the conditional probabilities $p(x_i|x_{\mathrm{pa}(i)})$ can be represented as conditional probability tables (CPTs). Note that the size of each table is the product of the sizes of the state spaces of the variables concerned. In order to represent genotypes in Equation 2 in a computationally efficient way, we use the following Markov representation.

7.1. Markov Genotype Representation

Some markers can have many alleles, so the CPTs for genotypes become large. This can be avoided by representing genotypes by vectors of allele counts: $n_{ia} = 0, 1$, or 2, for alleles $a = 1, 2, \ldots, A$, for individual i. We assume that the reference population is in Hardy–Weinberg equilibrium, so the two alleles of an individual are chosen at random with allele frequencies $(q_1, \ldots, q_A)$. For an individual drawn from the gene pool,

$$(n_{ia})_a^A \sim \text{Multinomial}(2, (q_a)_a^A),$$

where q_a is the frequency of allele a for this marker in the relevant population.

Now, imagine the two alleles of a genotype being allocated sequentially. Let S_{ia} be the partial sums $S_{ia} = \sum_{b \le a} n_{ib}$. Then, $n_{i1} \sim \text{Bin}(2, q_1)$ and

$$n_{i,a+1} \mid (n_{i1}, n_{i2}, \ldots, n_{ia}) \sim \text{Bin}\left(2 - S_{ia}, q_{a+1} / \sum_{b>a} q_b\right).$$

Hence, a genotype is modeled by a Markov structure, and in this way computations can be done linearly in number of alleles. Thus, adding the partial sums of allele counts to a network allows the genotype to be modeled by a Markov structure, as displayed in **Figure 8**.

8. STATISTICAL MODEL FOR DNA MIXTURES

We now give details on a robust model for identification and separation of complex DNA mixtures from several contributors based on discrete data (allelic presence) and on continuous peak height data. We first give a brief summary of the main features of the DNA mixture model described by Cowell et al. (2015). The model is an extension of the gamma model developed by Cowell et al. (2007a) and Cowell et al. (2007b), and used by Cowell et al. (2011). This model takes fully into account the peak heights and the possible artifacts, such as stutter, dropin, dropout, and silent alleles, and contamination that might occur in the DNA amplification process. These artifacts are

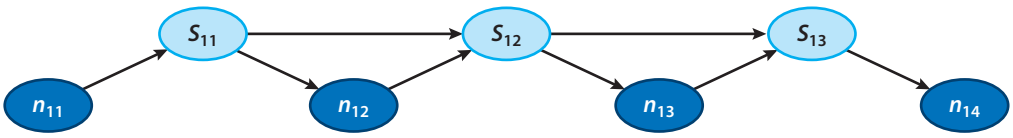


Figure 8

Markov genotype representation of one individual $i = 1$ for an allele taking $A = 4$ values. The dark blue nodes represent the allele counts n_{ia} and the light blue nodes represent the partial sums $S_{1a} = \sum_{b \le a} n_{1b}$.

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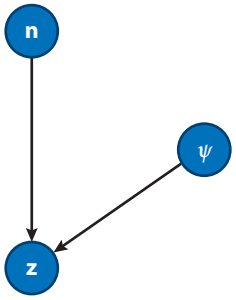


Figure 9

Directed acyclic graph (DAG) of the joint model for peak heights **z**, with parameters $\psi = (\phi, \mu, \sigma)$ and genotypes represented by allele counts **n**.

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often present when amplifying tiny amounts of DNA, like LTD. The computations exploit an efficient BN representation of the model.

We need to build a joint model for the peak heights **z** in the EPG (or EPGs), the genotypes, represented by allele counts **n** of the potential contributors to a mixture. In Section 10, we show how to extend this model to allow for inference about kinship with mixture contributors. The DAG in **Figure 9** shows the dependence structure between the peak heights **z**, the genotypes represented by allele counts **n**, and the parameters ψ of the mixture model.

The joint model is written in stages. In the first stage, genotypes are represented as allele counts. Given the genotypes, the second stage consists of modeling peak heights as continuous variables. Finally, the model can be generalized to allow inference about relationships, uncertainty in allele frequencies (UAF), heterogeneity in the population, and other issues.

8.1. Peak Height Model for Fixed Genotypes

The model assumes that the variability in peak height at an allelic position is independent of the variability at other allelic positions when the model parameters and genotypes are considered fully known. Consider allele a, the variability of the peak height Z_a at a can be expressed by a gamma distribution

$$Z_a \sim \Gamma\left\{\frac{1}{\sigma^2} D_a(\phi, \xi, \mathbf{n}), \mu\sigma^2\right\}, \qquad 5.$$

where ϕ_i denotes the proportion of DNA originating from individual i prior to PCR amplification; n_{ia}, $i = 1, \ldots I$; $a = 1, \ldots, A$ is the number of type a alleles for individual i; $D_a(\phi, \xi, \mathbf{n}) = (1 - \xi) \sum_i \phi_i n_{ia} + \xi \sum_i \phi_i n_{i,a+1}$ are the effective allele counts after stutter; and in a sample from a single donor, where no dropout or stutter has occurred, μ is the mean peak height and σ is the coefficient of variation. For example, $\sigma = 0.58$ corresponds to the standard deviation of the peak being 58% of its mean μ. The model takes into account artifacts: stutter, whereby a proportion of a peak belonging to allele a appears as a peak at allele $a - 1$, and dropout, when alleles are not observed because the peak height is below a detection threshold C. The back-stutter parameter ξ determines the mean proportion of stutter that may be observed in the allelic position one repeat less. Here ξ is the ratio of the stutter peak with respect to the parent plus the stutter peak, rather than the more commonly used ratio between the stutter peak and the parent peak (Butler 2005). Other distributions for the peak heights Z_a, like the log-normal or Weibull, might be equally well suited.

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The evidence E consists of the peak heights $\mathbf{z}$ as observed in the EPGs, as well as the genotypes of known individuals. For given genotypes of the contributors, expressed as allele counts $\mathbf{n} = (n_{ia}, i = 1, \ldots I; a = 1, \ldots, A)$, given proportions ϕ, and given values of the parameters $\psi = (\mu, \xi, \sigma)$, all observed peak heights are independent and, for a given hypothesis $\mathcal{H}$, the full likelihood is obtained by summing over all possible combinations of genotypes $\mathbf{n}$ with probabilities $P(\mathbf{n} \mid \mathcal{H})$ associated with $\mathcal{H}$:

$$L(\mathcal{H}) = \Pr(E \mid \mathcal{H}) = \sum_{\mathbf{n}} L(\psi \mid \mathbf{z}, \mathbf{n}) P(\mathbf{n} \mid \mathcal{H}), \qquad 6.$$

where

$$L(\psi \mid \mathbf{z}, \mathbf{n}) = \prod_m \prod_a L_{ma}(z_{ma})$$

and

$$L_{ma}(z_{ma}) = \begin{cases} g\{z_{ma}; \sigma^{-2} D_a(\phi, \xi, \mathbf{n}), \mu\sigma^2\} & \text{if } z_{ma} \geq C \\ G\{C; \sigma^{-2} D_a(\phi, \xi, \mathbf{n}), \mu\sigma^2\} & \text{otherwise,} \end{cases} \qquad 7.$$

with g and G denoting the gamma density and cumulative distribution function, respectively.

The number of terms in this sum is huge for a hypothesis that potentially involves several unknown contributors to the mixture. For example, for one unknown contributor in a DNA mixture calculated on $M = 13$ markers, each with $A = 20$ possible alleles, there are roughly 1,030 terms in the sum in Equation 6. However, Equation 6 can be calculated efficiently by BN techniques that represent the genotypes using a Markovian structure, in which the allele counts for each individual are modeled sequentially over the alleles. The maximum likelihood estimate (MLE) parameters are obtained using the R package `DNAmixtures` (Graversen 2013), which interfaces to the `Hugin` application programming interface (**https://www.hugin.com/index.php/hugin-api/**) through the R package `RHugin` (Konis 2014). The software `EuroForMix` (Bleka et al. 2016) is based on this gamma model.

8.2. Computation

As mentioned in Section 8, the main computational challenge is associated with the marginalization over all unknown DNA profiles needed for evaluating the likelihood function. This is particularly challenging when the number of contributors increases—as the number of terms in Equation 6 becomes intractably huge, so that brute force summation is infeasible. To enable efficient computation of the sum, we use an appropriate BN, as in **Figure 8**, to represent the genotypes of the unknown contributors and the observations. As markers are independent for fixed model parameters, the model can be represented by a separate BN for each marker. Here we only briefly describe the basic structure of these networks and refer the reader to Graversen & Lauritzen (2015) for further details on the networks and associated computational issues.

8.2.1. Including peak height information. The exact probability propagation methods of BNs work only for discrete variables and a very specific class of models with some continuous nodes (continuous Gaussian models). So how can we model continuous data, such as EPG peak heights, in a BN? The probability propagation algorithm actually computes expectations of products of functions of margins of $p(x)$, where $p(x) = \prod_{i \in V} p(x_i \mid x_{\mathrm{pa}(i)})$.

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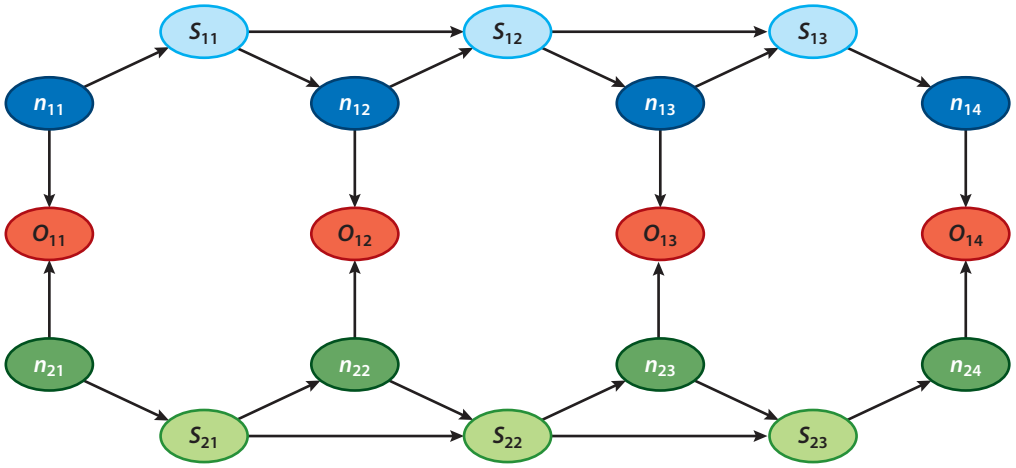


Figure 10

Bayesian network for a marker with $a = 1, \ldots, 4$ possible allelic types, modeling the genotypes as allele counts of 2 unknown contributors $i = 1$ in blue and $i = 2$ in green. The red nodes are the auxiliary variables O_a conditioning on the observed peak heights z_a.

This can be achieved by introducing a dummy binary variable Y_A into the BN for each such factor. In particular, the likelihood in Equation 6 is an expectation over the discrete variables $P(\mathbf{n} \mid \mathcal{H})$, so probability propagation (Lauritzen & Spiegelhalter 1988) is ideally suited to this computation.

The distribution of the peak height at allele a depends on both n_{ia} and $n_{i,a+1}$ for both known and unknown contributors i. To enable simple computation of the terms in the likelihood function (Equation 6), we represent the model as a BN shown in **Figure 10**.

The information about peak height observations is incorporated via binary nodes O_a in **Figure 10**; these represent whether a peak for allele a has been seen. By suitable specification of the conditional distribution of O_a given its parent nodes, conditioning on O_a being true or false will correspond to conditioning on the observed peak heights z_a; this enables, in particular, fast evaluation of the likelihood function by a single propagation in the BN (Graversen & Lauritzen 2015).

The Markov representation of the genotypes themselves ensures the complexity grows linearly with the number of allelic types A rather than polynomially, as with genotypes, which have $A(A + 1)/2$ values.

9. DATABASES AND SENSITIVITY TO PRIOR ASSUMPTIONS

The evidentiary value of observing an allele depends on how common it is. It is thus necessary to have appropriate databases of allele frequencies in order to analyze forensic evidence. A core issue is that all existing and potential forensic databases have fundamental weaknesses. All are based on approaches to sampling and data analysis that cannot handle the concept of systematic differences in nonhomogeneous populations, nor can they properly handle issues associated with systematic sampling and measurement errors. Ideally, the conclusions that can be validly drawn must take full account of the inherent uncertainty associated with the database itself.

There are also criminal databases that contain the DNA profiles of offenders. An offender will be placed in the crime database according to the laws of each country. Furthermore, different countries use different sets of markers to build their crime databases.

The allele frequencies assumed when conducting probabilistic forensic inference against an assumed background population are not fixed probabilities, but empirical frequencies in a database.

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Table 3 Example of the likelihood ratios (LRs) for comparing the prosecution and defense hypotheses when using different reference populations

Reference population	Italian	Macedonian Romani	Portuguese Romani	Eastern Slovakian Romani
LR	24.16	9.38	2.91	19.94

An imperfect idealization is to regard these databases as independently drawn random samples from corresponding populations. Green & Mortera (2009) develop a model for incorporating uncertain allele frequencies (UAF), population heterogeneity (HET), and kinship corrections based on the possibility of alleles that are IBD. As noted by Green & Mortera (2009), the simple kinship correction used by Balding (2013) ignores the dependence between markers that are created by identity by descent or by population heterogeneity, which may generally still lead to an overstatement of the weight of evidence.

Within a population, the various alleles of STR markers do not occur equally often; some can be quite common and some quite rare. When carrying out probability calculations based on DNA, forensic scientists use estimates of probabilities based on allele frequencies in profiles of a sample of individuals in a reference database. The sample sizes typically range from a few hundred to thousands of individuals. For example, the NIST STR database (**https://strbase.nist.gov/NISTpop.htm**) presents tables of US population STR-allele frequencies for Caucasians, African Americans, and Hispanics based on sample sizes of 361, 342, and 236 individuals.

It is important in any forensic DNA analysis, be it single source or a DNA mixture case, to use the appropriate reference population database and to account for the size of the database. This was highlighted in a recent case study concerning a series of mixed DNA profiles (Dotto et al. 2019). The case involved several different persons of interest of a small, inbred population of Romani origin. The results showed how the use of different population database allele frequencies for analyzing the DNA mixtures can lead to very different results, some seemingly inculpatory and some seemingly exculpatory. **Table 3** shows an example of the LRs for comparing the prosecution and defense hypotheses when using different reference populations. The LRs for the Italian Caucasian reference population and Eastern Slovakian Romani reference population are much higher than those for the Macedonian and Portuguese Romani reference populations. This illustrates that the use of an incorrect reference population database of allele frequencies can lead to the incorrect identification of a DNA contributor.

When using peak information in DNA mixtures, the markers will no longer be independent, as in single-source analysis, because of the uncertainty in the shared latent variable ψ, which represents the proportion of DNA each contributor gives to the mixture. When the markers are independent—the baseline scenario—the LRs calculated for each marker can be simply multiplied. This also holds when modeling the uncertainty in the allele frequencies.

Table 4 shows an example, based on 6 markers, of the sensitivity in the log-likelihood ratios $\log_{10} LR$s (when comparing the prosecution hypothesis $H_p : v\&s$ to a defense hypothesis

Table 4 Example of the sensitivity in the log-likelihood ratios $\log_{10} LR$s for $H_p : v\&s$ versus $H_d : v\&U$, in a DNA mixture for the baseline scenario, UAF, and IBD, based on 6 markers

	Baseline	UAF	IBD
Exact	8.23	7.93	6.2
Product rule	6.75	6.44	6.46

Abbreviations: H_d, defense hypothesis; H_p, prosecution hypothesis; IBD, identity by descent; s, suspect; U, unknown; UAF, uncertainty in allele frequencies; v, victim.

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Table 5 Example of the sensitivity in the LRs in a DNA mixture for the three markers, D3S1358, vWA and FGA, when comparing H_p : $v\&s$ versus H_d : $v\&U$, when the suspect and unknown are unrelated and when they are closely related

	Markers		
	D3S1358	**vWA**	**FGA**
Unrelated	50.90	11.52	14.61
Parent-child	7.12	2.94	2.94
Half-siblings	12.49	4.69	4.89

The analysis is made under the UAF scenario. Abbreviations: H_d, defense hypothesis; H_p, prosecution hypothesis; LR, likelihood ratio; s, suspect; U, unknown; UAF, uncertainty in allele frequencies; v, victim.

H_d : $v\&U$) in a two-person DNA mixture for the baseline scenario, UAF, and IBD. Note that when not considering the possibility of UAF or IBD, the LR against the defendant is overstated—it is more incriminating for a defendant, which could lead to a false incrimination. In this example, the across-marker results using the product rule give smaller LRs. Furthermore, the inference is much more certain when using peak height information. When only the discrete allele information is used, the overall $\log_{10} LR$ is 4.40 in baseline case.

Table 5 shows an example of the sensitivity in the LRs in a DNA mixture when comparing H_p : $v\&s$ versus H_d : $v\&U$, when the suspect and unknown are unrelated and when they are closely related. The analysis is made under the UAF scenario.

The closer the relationship between the suspect and the alternative unknown suspect, the smaller the LR becomes.

10. INFERENCE ABOUT RELATIONSHIPS FROM DNA MIXTURES

We now present methods for inference about relationships between contributors to a DNA mixture and other individuals of known genotype. The evidence for such a relationship is evaluated as the LR for the specified relationship versus the alternative that there is no relationship.

Green & Mortera (2017) extend the peak height model to allow for inference about the relationships between contributors to a DNA mixture with unknown genotype and other individuals of known genotype: A basic example would be testing whether a contributor to a mixture is the father of a child of known genotype (or indeed the similar question with the roles of parent and child reversed). Following commonly accepted practice, the evidence for such a relationship is presented as the LR for the specified relationship versus the baseline, null hypothesis, that there is no relationship at all, so the father is taken to be a random member of the population. Our methods are based on the statistical model for DNA mixtures of Cowell et al. (2015), in which a BN is used as a computational device for efficiently computing likelihoods and builds on that approach, but makes more explicit use of the BN in the modeling. The methods are not limited to particular details of the genotyping kits, allele frequencies, number of contributors, or hypotheses in these examples.

Further questions that can be answered by a similar approach include:

- Is a contributor to a mixture the brother of an individual of known genotype?
- Is a contributor to one mixture also a contributor to another mixture?
- Is a contributor to one mixture a brother of a contributor to another mixture?
- Is an individual of known genotype a relative of two contributors to a mixture who are mother and child?

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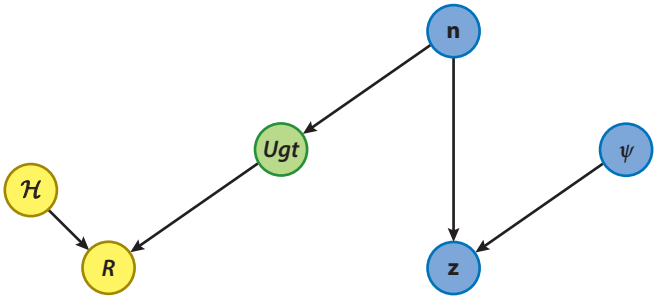


Figure 11

A DAG pictorial representation for establishing a potential relationship with a contributor to the mixture. The blue nodes represent the gamma model of Cowell et al. (2015). The yellow nodes denote the putative relationship between the mixture contributor Ugt and relatives with genotypes R, under the control of the hypothesis $\mathcal{H}$. Abbreviation: DAG, directed acyclic graph.

Mortera et al. (2016) gave an approximate method where only the most probable genotype of a mixture contributor is used to specify the questioned relationship, whereas Green & Mortera (2017) take all uncertainty about the mixture contributors into account.

The evidence for the relationship is evaluated as the LR for $\mathcal{H}_p$, which claims a specified relationship between a typed individual and an unknown contributor, versus the alternative $\mathcal{H}_0$, which claims that the unknown contributor to the mixture is a random member of the population. Let R denote the genotypes of the measured individuals, and let $U_i = U$ be a specified contributor to the mixture. The evidence $E = \{R, \mathbf{z}\}$ for which to evaluate the likelihood now consists of both the peak height information (quantified through $\mathbf{z}$) in the EPG(s) and the measured genotypes. The LR is

$$LR = \frac{\mathrm{P}(E|\mathcal{H}_p)}{\mathrm{P}(E|\mathcal{H}_0)},$$

where the hypotheses are now about potential relationships. The network in **Figure 11** represents the probabilistic dependence of the peak heights $\mathbf{z}$ on the allele counts $\mathbf{n}$ for the unknown contributors to the mixture, and the parameters $\psi = (\phi, \rho, \xi, \eta)$ of the gamma model. This dependence is represented in the right-hand part of the DAG in **Figure 11**.

Figure 12 shows how evidence can be combined under the hypothesis that a specific person whose genotype is among those measured in R is a parent of U_1. Note that, given the genotype of U_1, the genotypes in R are conditionally independent of the peak heights, $\mathbf{z}$.

Assuming that the model parameters ψ are the same under $\mathcal{H}_p$ and $\mathcal{H}_0$, the LR can be written as

$$LR = \sum_{Ugt} LR_{Ugt} \times \mathrm{P}(Ugt|\mathbf{z}), \qquad 8.$$

where $\mathrm{P}(Ugt|\mathbf{z})$ is the posterior probability of U_1 (the major unknown contributor) having genotype Ugt after taking into account the peak height information, and $\mathrm{LR}_{Ugt} = \mathrm{P}(R|Ugt, \mathcal{H}_p)/\mathrm{P}(R|\mathcal{H}_0)$ is the LR conditional on U_1 having genotype Ugt. For a specific parent genotype in R, pgt, this conditional LR is given as

$$LR_{Ugt} = \mathrm{P}(pgt|Ugt, \mathcal{H}_\mathrm{p})/\mathrm{P}(pgt|\mathcal{H}_0) = \begin{cases} n_{ia}/2q_a & \text{if } pgt = \{a,a\} \\ n_{ia}/4q_a + n_{ib}/4q_b & \text{if } pgt = \{a,b\}, \quad a \neq b, \end{cases} \qquad 9.$$

where q_a and q_b are the allele frequencies in a specific reference database, and n_{ia} and n_{ib} are the allele counts for Ugt.

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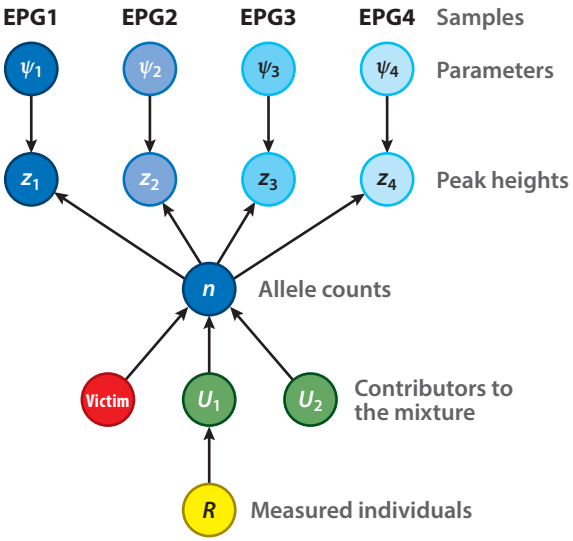


Figure 12

A DAG pictorial representation for establishing a potential relationship with a contributor to the mixture. The blue nodes represent the gamma model of Cowell et al. (2015). Here we assume a total of at most three contributors, the victim and two unknowns U_1 and U_2, to the four crime scene samples (EPGs 1–4), each having their own set of model parameters $\psi_i, i = 1, \ldots, 4$. The potential relationship is between the unknown mixture contributor U_1 and relatives with genotypes in R. Abbreviation: DAG, directed acyclic graph.

In order to compute the LR (Equation 8), four different methods were developed. A first method, termed weighted LR, uses the distribution of a contributor's genotype obtained from the mixture deconvolution and then computes the LR algebraically. A second method, termed additional likelihood nodes, is a modification of the Cowell et al. (2015) model, incorporating one or more additional auxiliary variables based on the relationship in question, as shown in **Figure 13**.

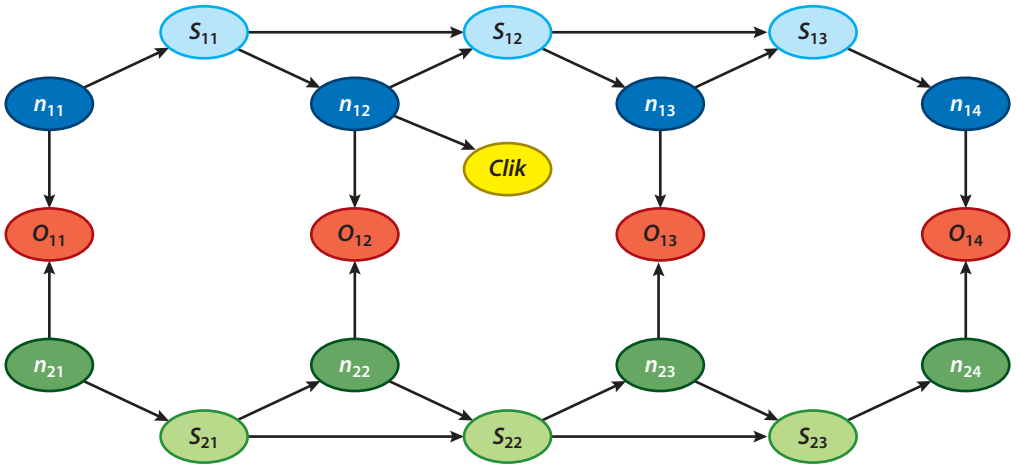


Figure 13

Bayesian network representation for an application of the additional likelihood nodes (ALN) method, showing the additional child likelihood node *Clik* for a case where there are $A = 4$ alleles and the parent genotype $pgt = \{2, 2\}$.

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The alternative methods are described in section 13 of Green & Mortera (2017). The calculations can be performed in R using a suite of functions `KinMix` (available at **https://petergreenweb.wordpress.com/kinmix/**) that extends the functionality of `DNAmixtures`.

Kaur et al. (2016) presented a method to handle a paternity relationship based on DNA mixtures that does not consider the peak height information but does allow a simple mutational event. This method is extended by Dørum et al. (2017) to allow for allelic dropout and dropin. Chung et al. (2010) describe familial searching on mixtures based solely on detected alleles and not accounting for artifacts. Slooten (2016) only considers sets of detected alleles for familial searching on DNA mixtures with dropout. The commercial software `STRmix` claims to implement familial searches against a database, for close relatives of contributors to mixed DNA. Testing whether mixtures have a donor in common or whether a relative of one mixture donor is a donor to a second mixture has been investigated by Ryan et al. (2016) and Slooten (2017). All these papers deal with simpler scenarios than those analyzed by Green & Mortera (2017); few of them use peak height information at all, and none of those that do incorporate it fully coherently through a probabilistic model.

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11. COMBINING EVIDENCE FROM MULTIPLE SAMPLES OR DIFFERENT MARKER SYSTEMS

As in the article by Graversen et al. (2019), there can often be multiple crime scene profiles available in a case. These either are replicates or originate from different samples. Furthermore, the samples can also be typed with different kits. There are many reasons for combining evidence, and an important one is that it strengthens the information about the profiles of any shared contributors. Combining the information in multiple profiles requires a slightly more complex analysis than that of single DNA mixture profiles, since it is now necessary to make assumptions about which—if any—contributors may be in common. When combining replicates, it is natural to make an assumption that contributors are the same; however, when combining profiles from different samples, one needs carefully to consider whether there is perhaps only a partial overlap. However, once a hypothesis describing the contributors is formulated, the mathematical details in extending the model from one to multiple crime scene profiles are completely straightforward. One can assume, in fact, that conditionally on the DNA profiles of the entire pool of contributors (and the model parameters), the peak heights in one EPG are independent of the peak heights in the other EPGs.

Figure 12 shows how evidence can be combined under the hypothesis that a specific person whose genotype is among those measured in R is a parent of U_1.

12. LEAVING THE CONTRIBUTOR UNSPECIFIED, AND LIKELIHOOD RATIOS FOR UNIONS OF ALTERNATIVE HYPOTHESES

In the modeling of DNA mixtures, the contributors have to be labeled to ensure all parameters are identifiable; the convention used in `DNAmixtures` is for the contributors to be numbered from 1, in decreasing order of the estimated proportion they contribute to the mixture in the first trace.

In some situations, it is perfectly appropriate to formulate hypotheses about relationships in terms of the major contribution U_1. In other situations, we may prefer to assess a relationship with an unspecified contributor; this is formally a case of evaluating an alternative hypothesis $\mathcal{H}_p$ that is the union of two or more corresponding hypotheses about specified contributors, $\mathcal{H}_p = \mathcal{H}_1 \cup \mathcal{H}_2 \cup \cdots$, where $\mathcal{H}_k$ is the hypothesis that the relationship in question is with the kth contributor.

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The problem of defining a LR for $\mathcal{H}_p$ against $\mathcal{H}_0$, given the LR_k for $\mathcal{H}_k$ against $\mathcal{H}_0$, $k = 1, 2, \ldots$ is a generic one.

A possible generic solution is to take $LR = \max LR_k$; this is similar to standard practice in evaluating generalized LRs in regular parametric problems, but is probably unacceptable in judicial work. More satisfactory in the present context would be to follow the Bayesian interpretation of the LR. A simple application of Bayes' theorem gives

$$LR = \frac{\sum_k LR_k \mathrm{P}(\mathcal{H}_k)}{\sum_k \mathrm{P}(\mathcal{H}_k)} = \sum_k LR_k \mathrm{P}(\mathcal{H}_k|\mathcal{H}_p),$$

which depends on the relative prior probabilities for the alternative hypotheses, but not their absolute probabilities.

13. SOFTWARE

There are a variety of software systems for analyzing DNA mixtures. These raise problems of interpretability, transparency, and consistency. In particular there is a tension between open source software and propriety commercial code. This tension is felt strongly in the courtroom, now that the outcomes of commercial algorithms are being introduced in criminal proceedings as part of the evidence. A danger in some of the methods, such as RMNE, is the failure properly to evaluate and communicate uncertainty. It is essential to report the uncertainty in any analysis using, e.g., the LR. Models, software, and data used both for validation studies and in real cases should be made available for examination and should be explained clearly in the report that is presented in court. Black-box methods should be avoided.

Every statistical analysis, together with the algorithms used in the software that is presented to the court, should have a specification document clearly illustrating all the details, and this should be disclosed to all parties in a case. This is especially true for commercial code.

Given the problems with subjective interpretation of complex DNA mixtures, a number of researchers have developed computer programs that apply various algorithms to interpret complex mixtures. The programs clearly represent a major improvement over purely subjective interpretation. However, they still require scientific scrutiny to determine

- whether the methods are scientifically valid, which should include defining the limitations on their reliability (that is, the circumstances in which they may yield unreliable results);
- whether the software correctly implements the methods published; and
- the scientific validity of their methods and algorithms across a broad number of possible settings.

There are several probabilistic genotyping computer programs that apply various models to interpret DNA mixtures. Some programs use discrete or semicontinuous methods, which use the allele information, possibly in conjunction with probabilities of allelic dropout and dropin, while other programs also include the continuous peaks in varying ways. **Table 6** gives a summary of some of the main software programs available for analyzing DNA mixtures. These systems vary in the input data they use, the modeling assumptions they make, and, thus, the resulting analysis. Consequently, they could all be correct in their own terms, but such discrepancies could cause confusion in the delivery of a case for a judge or jury.

In the Potsdam murder case (Sections 2 and 4), we saw that the two commercial software programs for analyzing DNA mixtures, `TrueAllele` and `STRmix`, gave conflicting conclusions concerning the defendant. These two commercial software programs have the following costs: (*a*) An educational user license for `STRmix` costs $6,000, together with the compulsory annual

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Table 6 Summary of main software programs available and their characteristics

Software	Open source	Model	Allows relatedness	Does model checking	# of unknown
DNAmixtures[a] and KinMix[b]	Yes	Continuous	Yes	Yes	10
EuroForMix[c]	Yes	Continuous	Yes	No	?
likeLTD[d]	Yes	Semicontinuous	Yes	No	3
LRmix[e]	Yes	Discrete	No	No	3
Lab Retriever[f]	Yes	Discrete	No	No	4
LiRa[g]	For internal use only	Continuous	Yes	No	4
TrueAllele®[h]	No	Continuous	No	No	10
STRmix™[i]	No	Continuous	No	No	4

[a] **http://dnamixtures.r-forge.r-project.org/**. It requires Hugin.
[B] **https://petergreenweb.wordpress.com/kinmix/**.
[c] **http://www.euroformix.com/**.
[d] **https://CRAN.R-project.org/package=likeLTD**.
[e] **http://lrmixstudio.org/**.
[f] **https://github.com/SCIEG/LabRetriever**.
[g] **https://cdnmedia.eurofins.com/european-west/media/1418957/lgc_lira_fact_sheet_en_0815_90.pdf**.
[h] **https://www.cybgen.com/products/casework.shtml**.
[i] **https://www.strmix.com/**.

upgrade of $1,000 per year and full user training (an additional cost) for at least one user prior to purchase, and (*b*) the academic cost for TrueAllele depends on the project.

The probabilistic genotyping software programs clearly represent a major improvement over purely subjective interpretation and those that use fully continuous models are preferable, as they take all the information in the EPG into account. However, this software still requires validation studies to determine (*a*) whether the methods are scientifically valid and what the limits on their reliability are (for example, if they are still valid with LTD when only a few cells are amplified) and (*b*) whether the software correctly implements the models.

As stated in the President's Council of Advisors on Science and Technology report (PCAST 2016, p. 79) "appropriate evaluation of the proposed methods should consist of studies by multiple groups, not associated with the software developers, that investigate the performance and define the limitations of programs by testing them on a wide range of mixtures with different properties." Accordingly, the following issues need to be addressed:

- How well do the methods perform as a function of the number of contributors to the mixture?
- How do the methods perform when the mixtures include related individuals?
- How well do the methods perform as a function of the amounts of DNA?
- How do the methods deal with the artifacts like stutter, dropout, and dropin present in the amplification of DNA? Do they model them appropriately in a fully coherent way?
- Do the methods clean the data of the artifacts? This could cause problems if the cleaning is done invisibly according to preset criteria and a potential true allele could be eliminated.
- Do the methods perform model checking on the specific case analyzed?

14. DISCUSSION AND FURTHER DEVELOPMENTS

In both paternity and criminal cases, one sometimes needs to allow for dependence between genotypes of contributors. This possibility will obviously affect the evidential value of the mixture. We

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distinguish two cases: (*a*) populations with high relatedness due to inbreeding and (*b*) specific close relationships, e.g., father and son, or two or more brothers engaged in a joint criminal activity. In both cases, the genotypes of two or more actors will be positively associated through identity by descent, rather than being independent draws from the gene pool. The phenomenon is the same in both cases, but they require different modeling approaches.

14.1. Populations with High Relatedness

As shown by Green & Mortera (2009), various probabilistic models for small populations lead to the same Pólya urn scheme for repeated draws $g_1, g_2, g_3 \ldots$ of genes from a population,

$$g_1 \sim p, \quad g_{n+1}|g_1, g_2, \ldots, g_n \sim \frac{1}{n+\alpha}\sum_{i=1}^{n}\delta_{g_i} + \frac{\alpha}{n+\alpha}p,$$

where p is the gene pool distribution, and $\alpha = (1-\theta)/\theta$. Balding & Nichols (1995) discuss this problem and model comprehensively, and they identify θ with Sewell Wright's F_{ST}, a measure of interpopulation variation (Wright 1951). It can be interpreted in terms of kinship, the proportion of alleles that share a common ancestor; they propose what has become known as the θ correction, which is very widely used to adjust forensic match probabilities for kinship. As observed by Green & Mortera (2009), exactly the same probabilistic model for the joint distribution of multiple genes arises in a simple model for UAF, in which the true allele frequencies are treated as unknowns with a Dirichlet distribution and the database used for calculation is regarded as a multinomial sample from these true frequencies. The parameter α is then the effective size of the database. This model is amenable to implementation as a BN (Green & Mortera 2009).

In the discussion of Cowell et al. (2015), Green observes that when genotypes are represented sequentially over alleles the conditional distributions over alleles, can be expressed as Beta–Binomial conditional distributions.

14.2. Specific Close Relationships in Mixtures

Our current work with P.J. Green concerns modeling mixtures to test if contributors have specific close relationships, e.g., a mixture of two or more brothers. The general setting is to encode the joint distribution of the genotypes of specified individuals in a pedigree, in a form amenable to computational inference based on peak heights in a DNA mixture with these individuals as contributors. When this inference is to be implemented using BNs, a natural approach is to model the pedigree directly in a graphical model. This can then be extended to tests for relatedness with typed individuals, under assumptions of relatedness of contributors and between contributors and typed individuals who might be related.

We have shown that a wide range of problems concerning DNA mixtures, with extensions to combining different mixtures, allowing for relationship inference in DNA mixtures, and more, can be handled coherently. The modular approach based on BNs provides a tool kit for new problems of the kind shown here.

APPENDIX: BAYESIAN NETWORKS

In this appendix, aimed at the more technically-interested reader, we give more detail on DAGs and BNs. A BN is a form of DAG comprising a finite set V of nodes, with arrows between some of the nodes, in such a way that it is not possible, by following the arrows, to return to one's starting point. The nodes having arrows pointing out from them and into a node $v \in V$ are called

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its parents and are denoted by pa(v); those with arrows pointing into them out from v are its children. This terminology is extended, in an obvious way, to ancestors, descendants, and so on.

Independence Properties

Each node $v \in V$ is identified with random variables X_v, and for $S \subseteq V$ we write X_S for $(X_v : v \in S)$. A BN represents a joint distribution P over the variables $(X_v : v \in V)$ when it is the case that, under P, each node in V is independent of all its nondescendants, conditional on its parents

$$X_v \perp\!\!\!\perp X_{\mathrm{nd}(v)} \mid X_{\mathrm{pa}(v)}.$$

For example, in a distribution represented by **Figure 7**, the red node is independent of the yellow node, conditional on the green nodes.

Quantitative structure. Let $\mathcal{X}_v$ denote the set of states (possible values) of X_v. For $S \subseteq V$ we write $\mathcal{X}_S$ for $\times_{i \in S} \mathcal{X}_i$. For $x = (x_i : i \in V) \in \mathcal{X}_V$, its projection (restriction), $(x_i : i \in S)$, to $\mathcal{X}_S$ is denoted by x_S.

We specify, for each node $v \in V$, a distribution over $\mathcal{X}_v$, conditional on each configuration $x_{\mathrm{pa}(v)} \in \mathcal{X}_{\mathrm{pa}(v)}$ of the states of its parents: The associated (discrete or continuous) density is denoted by $p(x_v \mid x_{\mathrm{pa}(v)})$. For a distribution P represented by the BN, the joint density of all variables factorizes as the product of its parent-child conditional densities:

$$p(x) = \prod_{i \in V} p(x_i | x_{\mathrm{pa}(i)}).$$

Conversely, any distribution with a density of this form will be represented by the BN.

When the variables are discrete, the parent-child conditional probability distributions can be represented as tables of the conditional probabilities $p(x_v | x_{\mathrm{pa}(v)})$. The overall size of each table is the product of the sizes of the state spaces of the variables concerned.

Computation. The structure of a BN incorporates a degree of modularity, which makes it possible to execute complex computations by dividing them up into a sequence of local computations, each involving a subset of the variables, known as a clique. This is particularly useful when all cliques are relatively small. There exist elegant algorithms to identify the cliques and streamline the computations, and these have been implemented in a number of software packages. In particular, for a discrete distribution specified by its parent-child CPTs, the Lauritzen–Spiegelhalter probability propagation algorithm (Lauritzen & Spiegelhalter 1988) enables efficient computation of all marginal densities $\{p(x_i) : i \in V\}$. Moreover, for any $S \subseteq V$ and evidence $X_s = x_S^*$, essentially the same algorithm computes the prior probability of obtaining that evidence, $\mathrm{P}(X_S = x_S^*)$, and the marginal posterior densities $\{p(x_i \mid X_S = x_S^*) : i \in V\}$. It is also straightforward to compute revised probabilities after incorporating external likelihood evidence relating to a subset of the variables.

DISCLOSURE STATEMENT

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